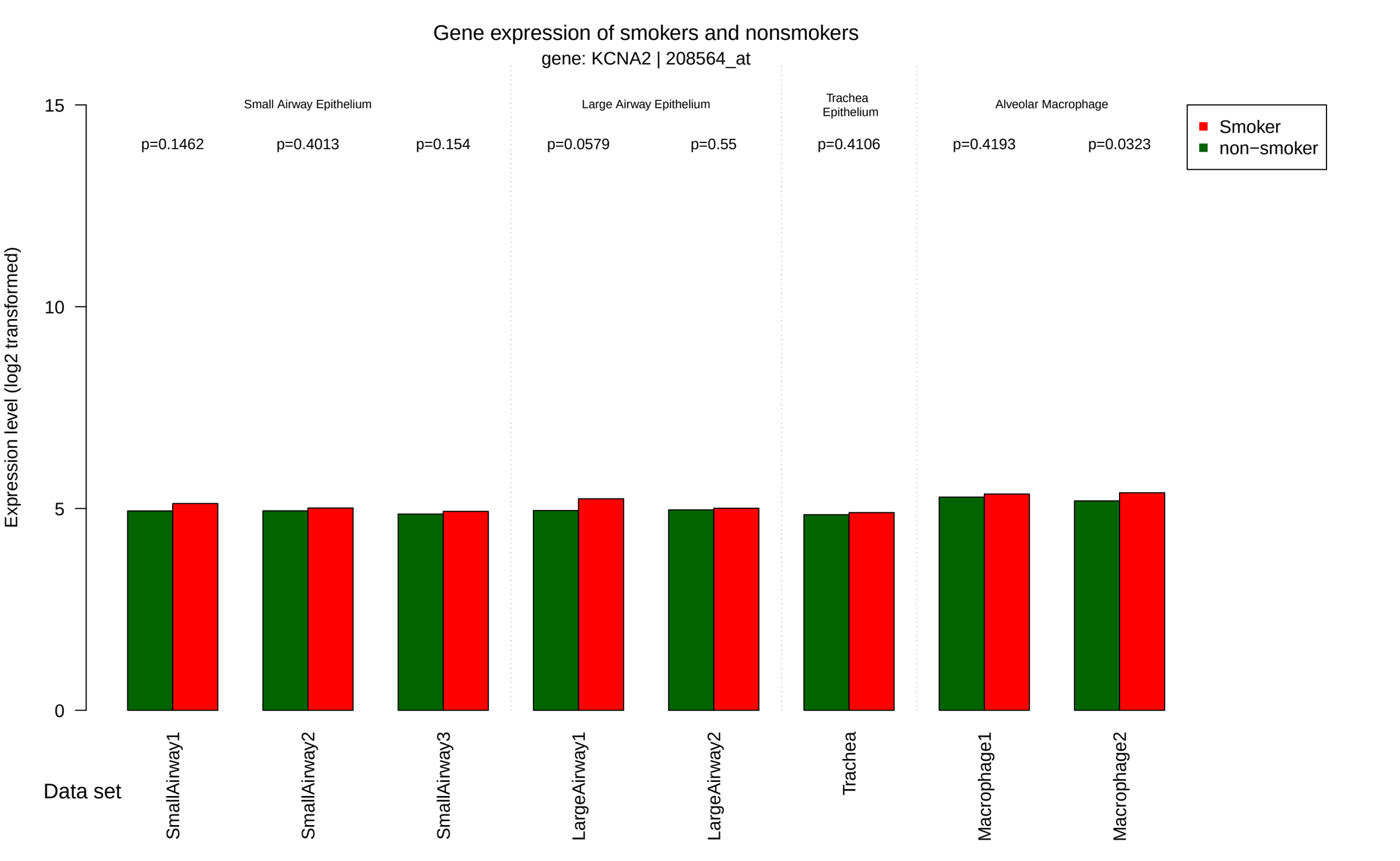
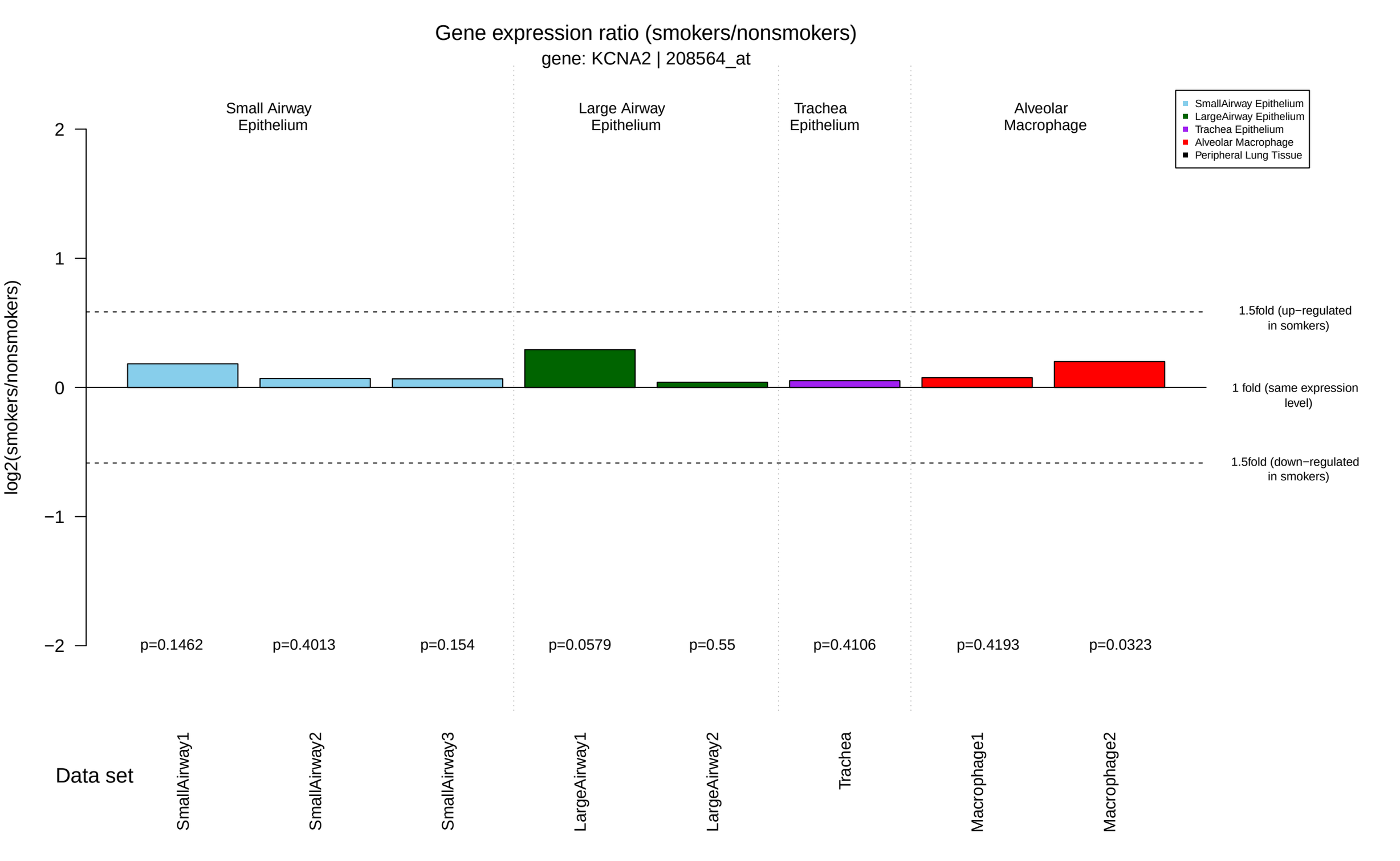
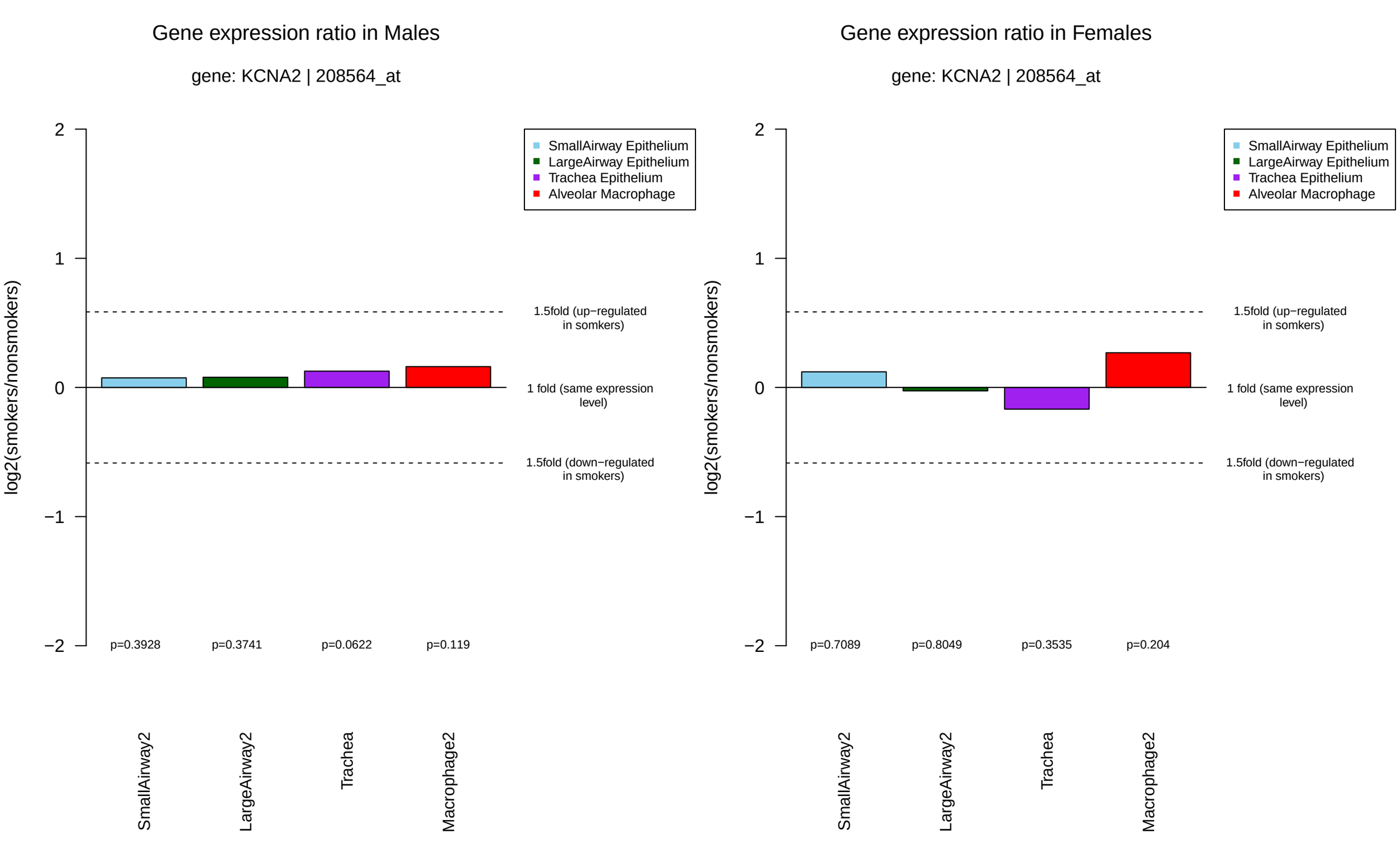
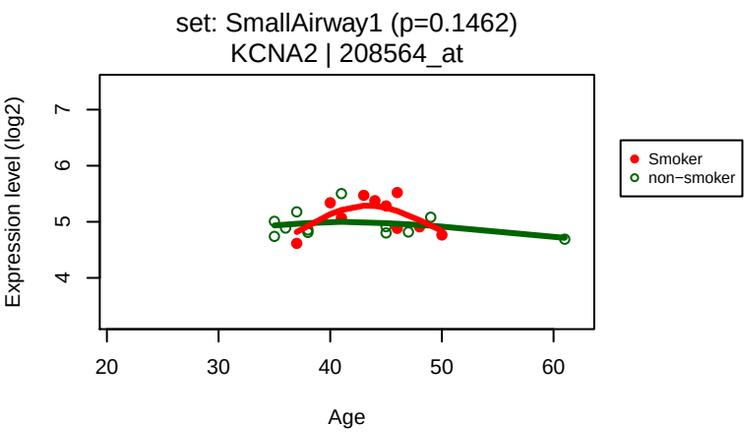
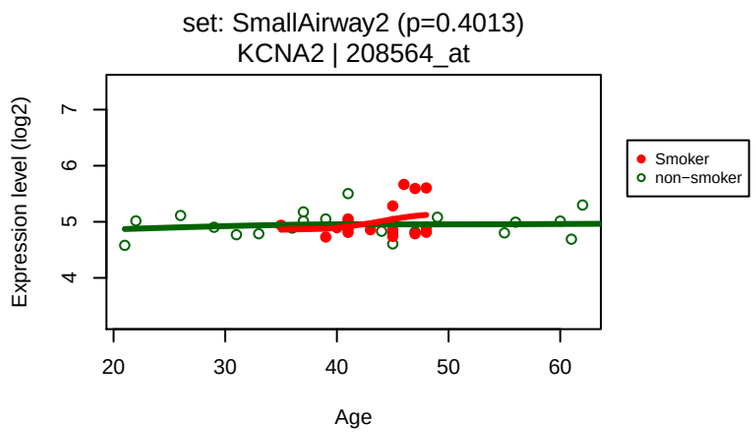
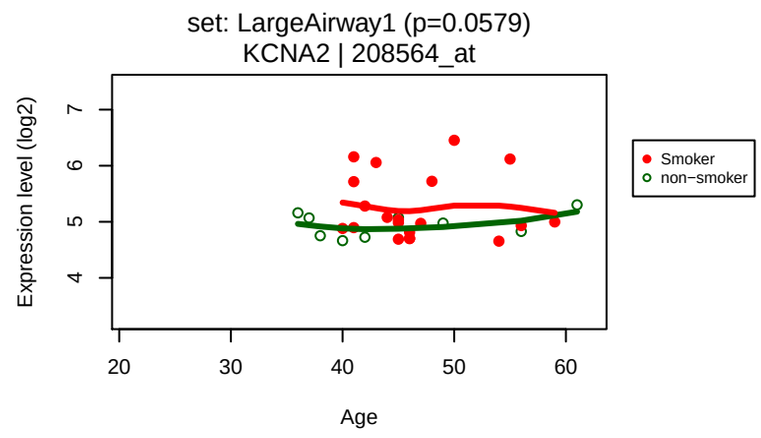

Gene expression of smokers and nonsmokers
gene: KCNA2 | 208564_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1462
p=0.4013
p=0.154
p=0.0579
p=0.55
p=0.4106
p=0.4193
p=0.0323
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

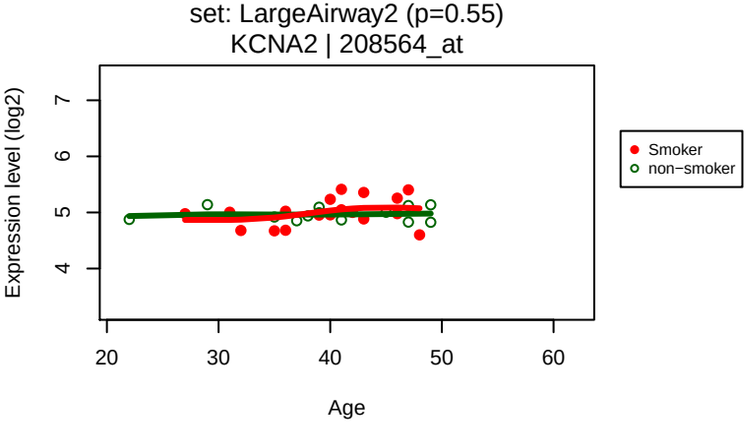

Gene expression ratio (smokers/nonsmokers)
gene: KCNA2 | 208564_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1462
p=0.4013
p=0.154
p=0.0579
p=0.55
p=0.4106
p=0.4193
p=0.0323
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

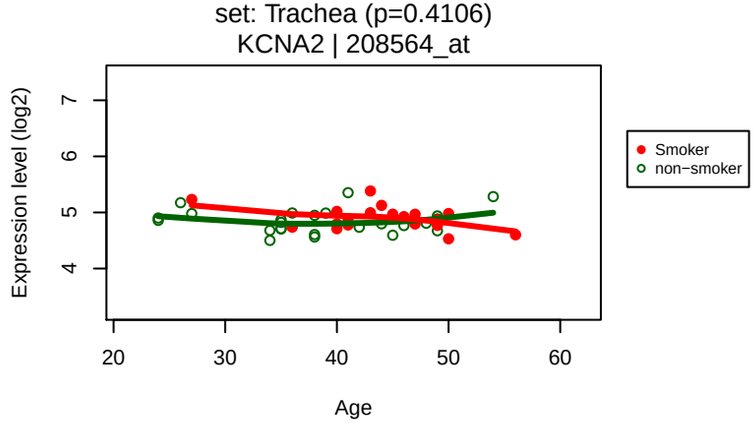

Gene expression ratio in Males
gene: KCNA2 | 208564_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3928
p=0.3741
p=0.0622
p=0.119
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNA2 | 208564_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7089
p=0.8049
p=0.3535
p=0.204
SmallAirway2
LargeAirway2
Trachea
Macrophage2

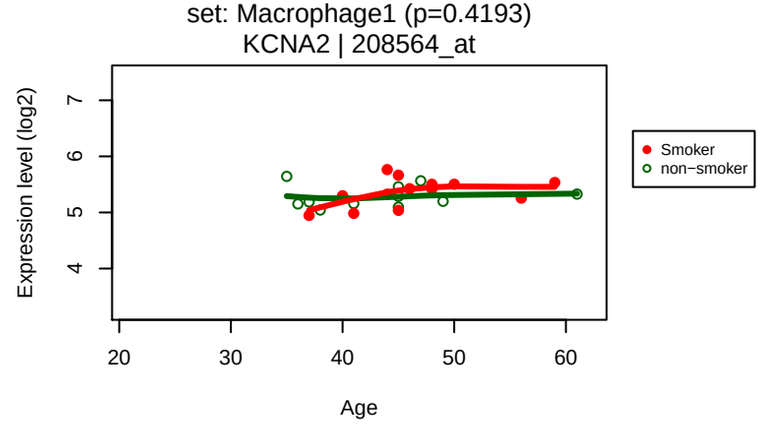

set: SmallAirway1 (p=0.1462)
KCNA2 | 208564_at
Expression level (log2)
Age
Smoker
non−smoker

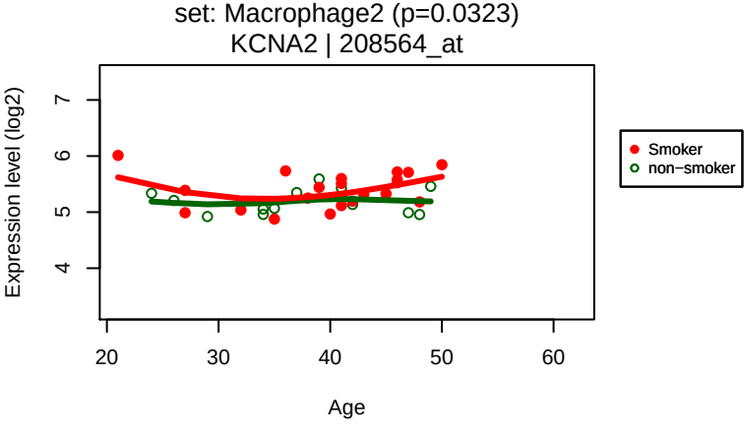

set: SmallAirway2 (p=0.4013)
KCNA2 | 208564_at
Expression level (log2)
Age
Smoker
non−smoker

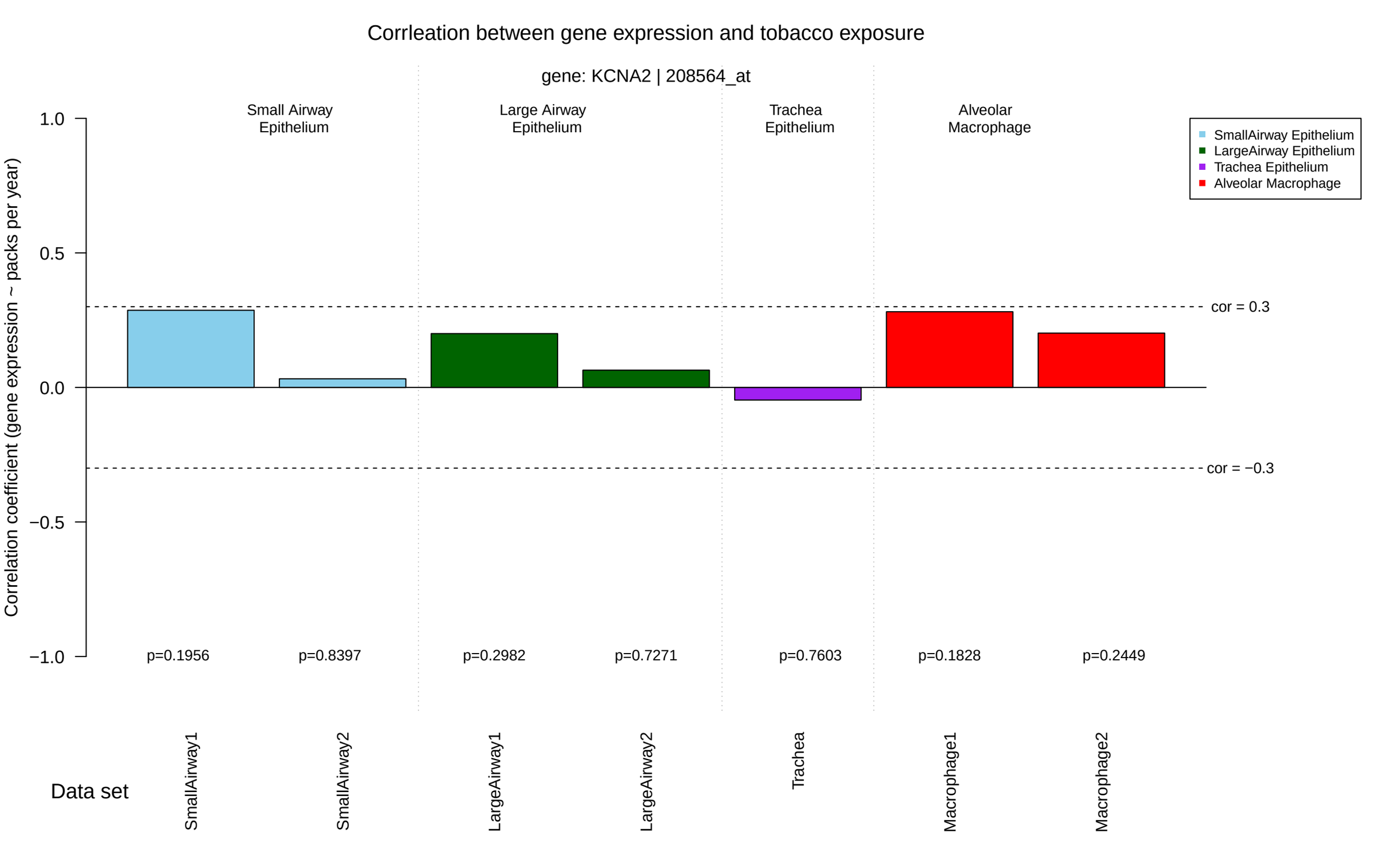

set: LargeAirway1 (p=0.0579)
KCNA2 | 208564_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.55)
KCNA2 | 208564_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4106)
KCNA2 | 208564_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4193)
KCNA2 | 208564_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0323)
KCNA2 | 208564_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNA2 | 208564_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1956
p=0.8397
p=0.2982
p=0.7271
p=0.7603
p=0.1828
p=0.2449
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2